Schematics For RK3566 BOX

RK3566_BOX_Demo1_DDR4P408DD4_V10

Main Functions Introduction

1) PMIC: DiscretePower(DCIN 12V)
2) RAM: DDR4 4x8Bit
3) ROM: eMMC,Nand Flash(Option)
4) Support: 1 x Micro SD Card3.0
5) Support: 1 x SATA3.0 Connector (7pin)
6) Support: 1 x USB2.0 OTG + 1 x USB3.0 HOST + 2 x USB2.0 HOST
7) Support: 1 x HDMI2.0 TX
8) Support: 1 x Optical S/PDIF TX
9) Support: 1 x 4Lane MIPI CSI RX
10) Support: 1 x 10/100/1000 Ethernet(RGMII)
11) Support: 1 x a/b/g/n/ac Wi-Fi + BT 5.0(2T2R)
12) Support: 1 x IR Receiver
13) Support: 1 x Recovery Key
14) Support: 1 x Power LED,1 x Working LED,1 x IR LED,1 x SATA ACT LED
15) Support: Debug UART

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Generate Bill of Materials

Header:

Item\tPart\tDescription\tPCB Footprint\tReference\tQuantity\tOption

Combined property string:

{Item}\t{Value}\t{Description}\t{PCB Footprint}\t{Reference}\t{Quantity}\t{Option}

Description

Note

Option

Notes

NOTE 1£°
Component parameter description
1. DNP stands for component not mounted temporarily
2. If Value or option is DNP, which means the area is reserved without being mounted

NOTE 2:

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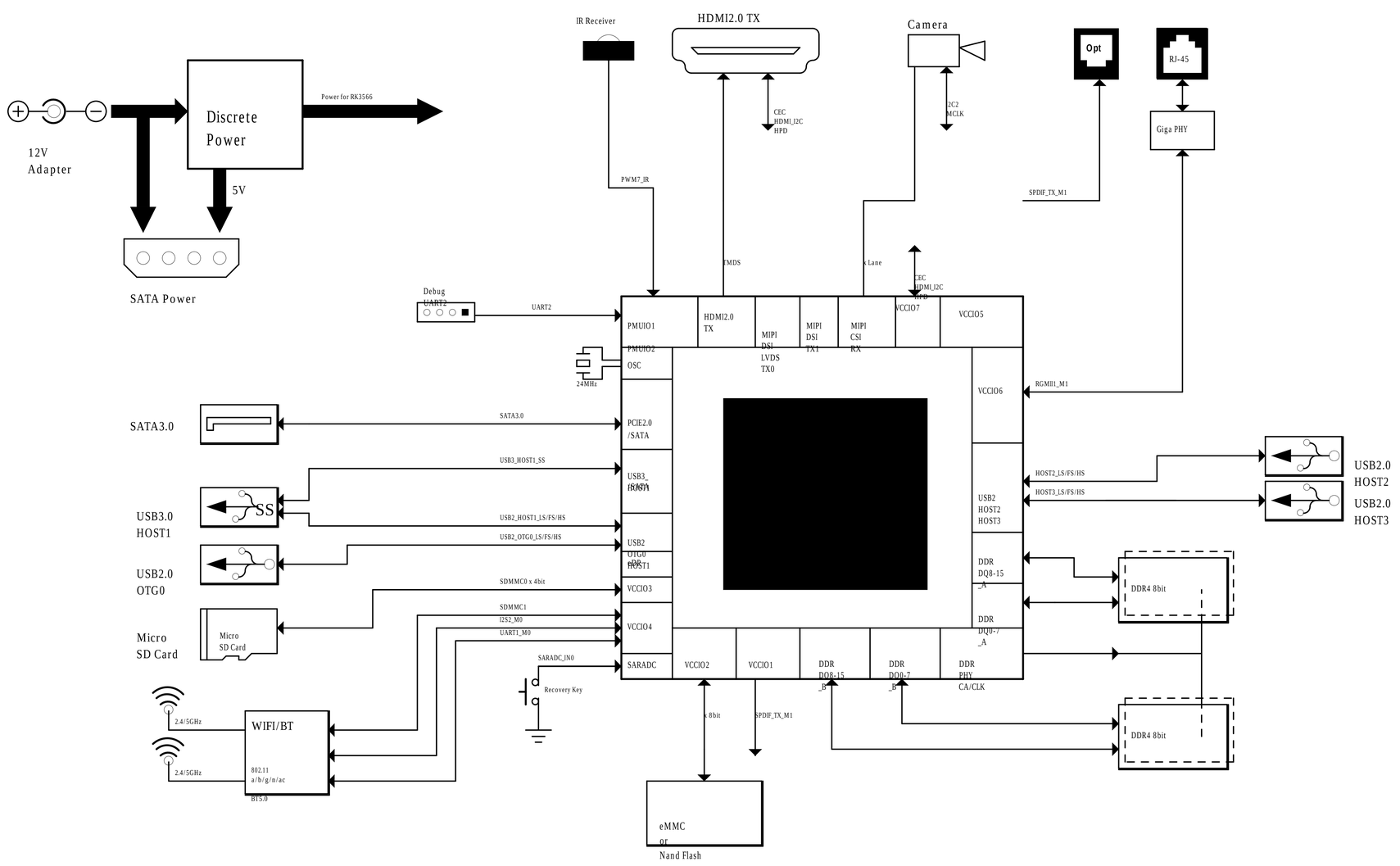
Revision History

Version	Date	By	Change Dsecription	Approved
V1.0	2021-01-05	Zhangdz	1:Revision preliminary version	
V1.1	2021-03-04	Zhangdz	1£°Add DDR4 8bit supports CS1/ODT1 2£°Change IR to PWM3_IR	

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RK3566 Ref Block Diagram

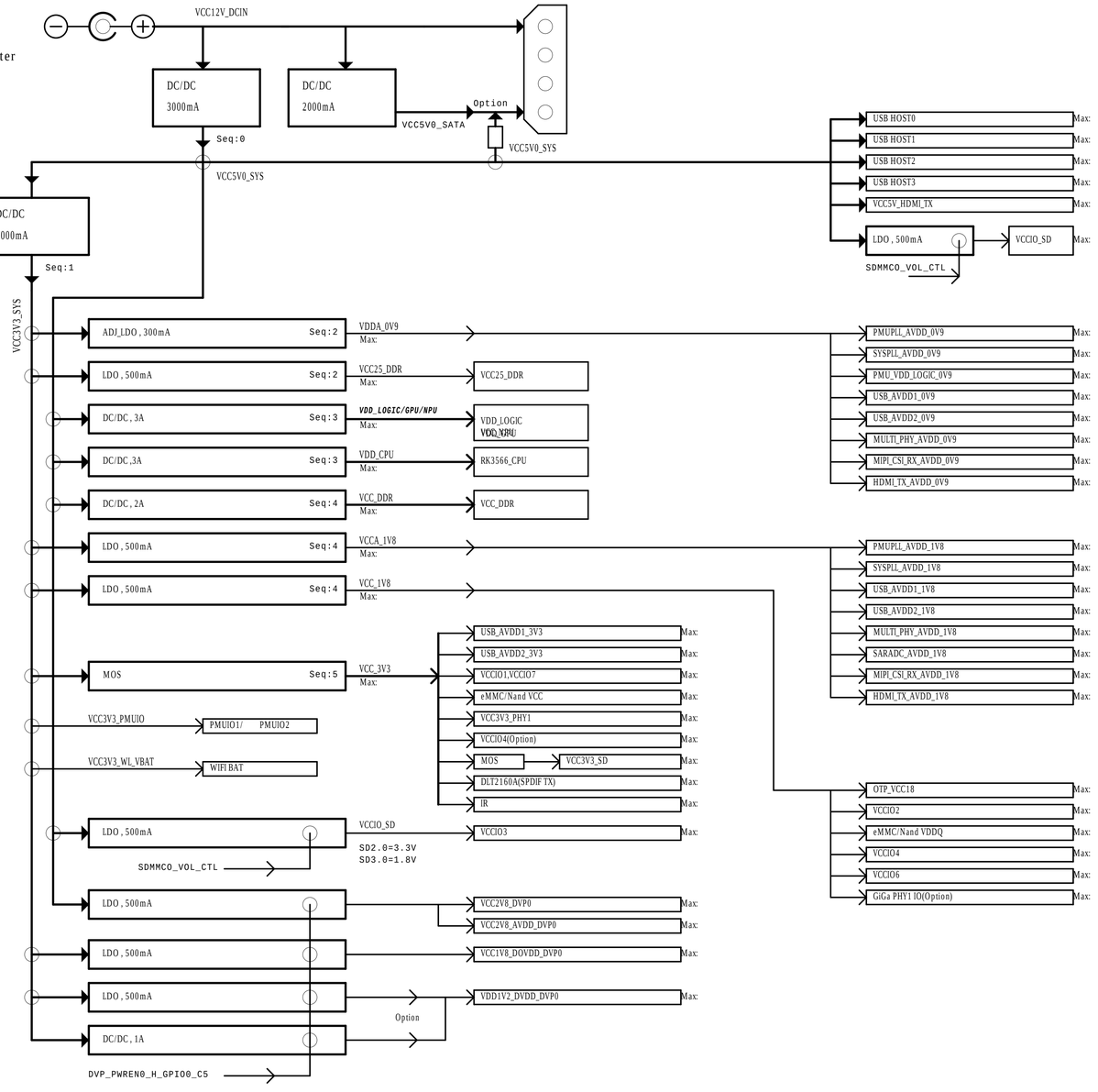


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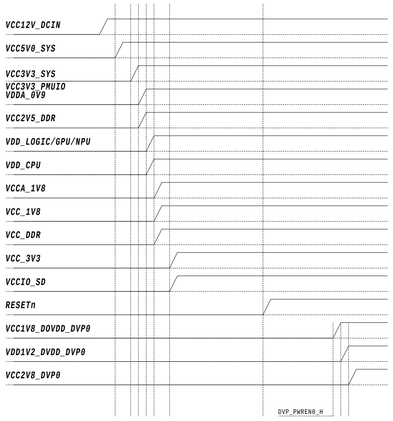
Power Diagram



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Power Sequence



Power Supply	Channel	Supply Limit	Power Net Name	Time Slot	Default Voltage
VCC12V_DCIN	BUCK	10A	VCC3V3_SYS	Slot:0	3.3V
VCC12V_DCIN	BUCK	3.0A	VCC3V3_SYS	Slot:1	5.2V
VCC3V3_SYS			VCC3V3_PMUIO	Slot:1	3.3V
VCC3V3_SYS	LDO	0.3A	VDDA_0V9	Slot:2	0.9V
VCC3V3_SYS	LDO	0.3A	VCC2V5_DDR	Slot:2	2.5V
VCC5V0_SYS	BUCK	3.0A	VDD_LOGIC/GPU/NPU	Slot:3	0.9V
VCC5V0_SYS	BUCK	5.0A	VDD_CPU	Slot:3	0.9V
VCC3V3_SYS	LDO	0.5A	VCC_1V8	Slot:4	1.8V
VCC3V3_SYS	LDO	0.5A	VCCA_1V8	Slot:4	1.8V
VCC5V0_SYS	BUCK	1.5A	VCC_DDR	Slot:4	1.2V
VCC3V3_SYS	MOS	2A	VCC_3V3	Slot:5	3.3V
VCC5V0_SYS	LDO	0.5A	VCCIO_SD	Slot:5A	3.3V
VCC3V3_PMUIO	RESETn				
VCC3V3_SYS	LDO	0.5A	VCC1V8_DOVDD_DVP0		1.8V
VCC3V3_SYS	LDO or BUCK	0.5A/1A	VDD1V2_DVDD_DVP0		1.2V
VCC5V0_SYS	LDO	0.5A	VCC2V8_DVP0		2.8V

IO Power Domain Map
Updates must be Revision accordingly!

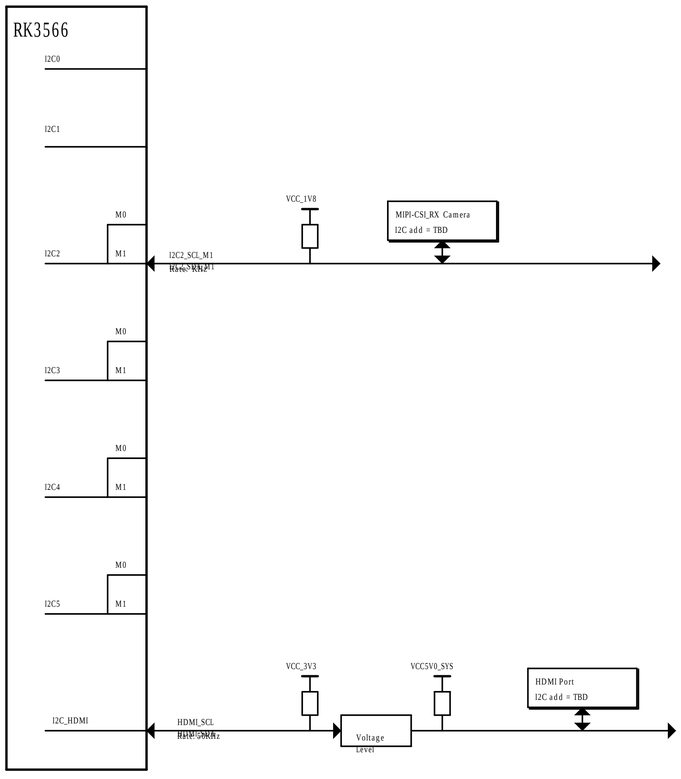
IO Domain	Pin Num	Support IO Voltage 3.3V	Support IO Voltage 1.8V	Actual assigned IO Domain Voltage Supply Power Net Name	Actual assigned IO Domain Voltage Power Source	Actual assigned IO Domain Voltage Voltage	Notes
PMUIO1	Pin Y20	✓	✗	VCC3V3_PMUIO	VCC3V3_SYS	3.3V	
PMUIO2	Pin W19	✓	✓	VCC3V3_PMUIO	VCC3V3_SYS	3.3V	
VCCIO1	Pin H17	✓	✓	VCC_3V3	VCC_3V3	3.3V	
VCCIO2	Pin H18	✓	✓	VCCIO_FLASH	VCC_1V8	1.8V	PIN "FLASH_VOL_SEL" must be logic High if VCCIO_FLASH=3.3V,FLASH_VOL_SEL must be logic low
VCCIO3	Pin L22	✓	✓	VCCIO_SD	VCCIO_SD	3.3V	SD2.0=3.3V-->SD3.0=1.8V
VCCIO4	Pin J21	✓	✓	VCCIO_WL	VCC_1V8	1.8V	
VCCIO5	Pin V10 Pin V11	✓	✓				
VCCIO6	Pin R9 Pin U9	✓	✓	VCC_1V8	VCC_1V8	1.8V	
VCCIO7	Pin V12	✓	✓	VCC_3V3	VCC_3V3	3.3V	

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I2C MAP

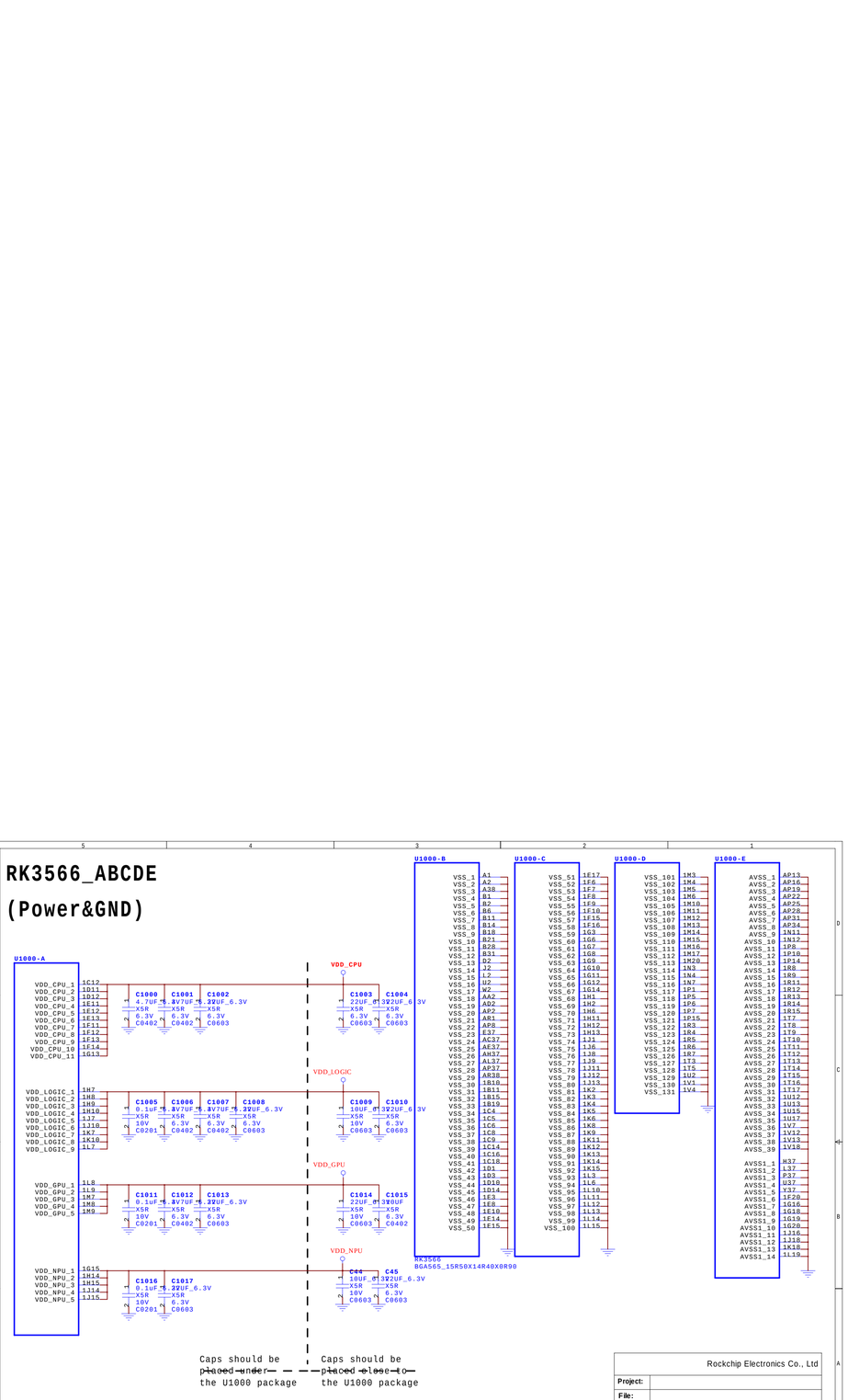


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RK3566_ABCDE
(Power&GND)

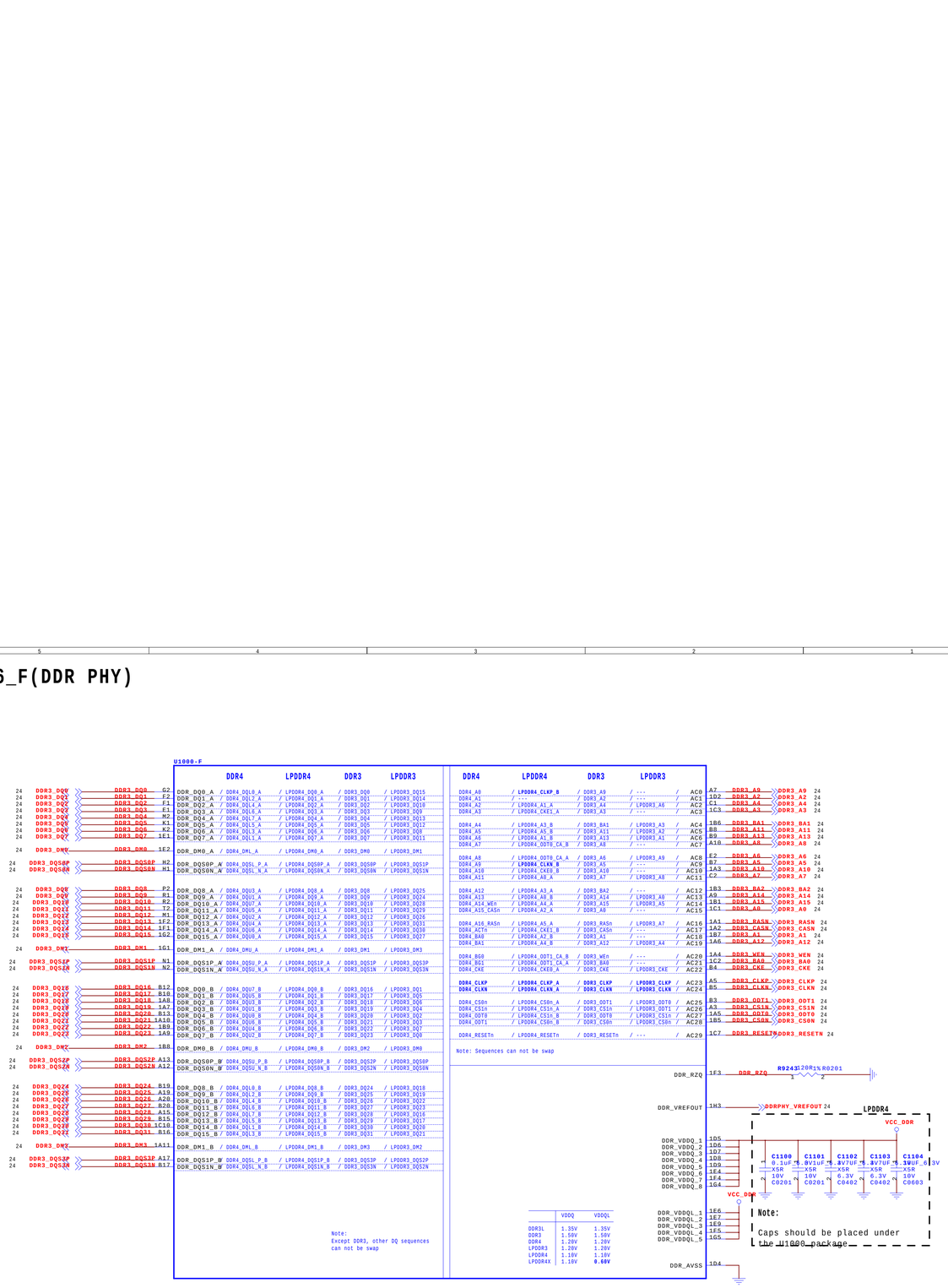


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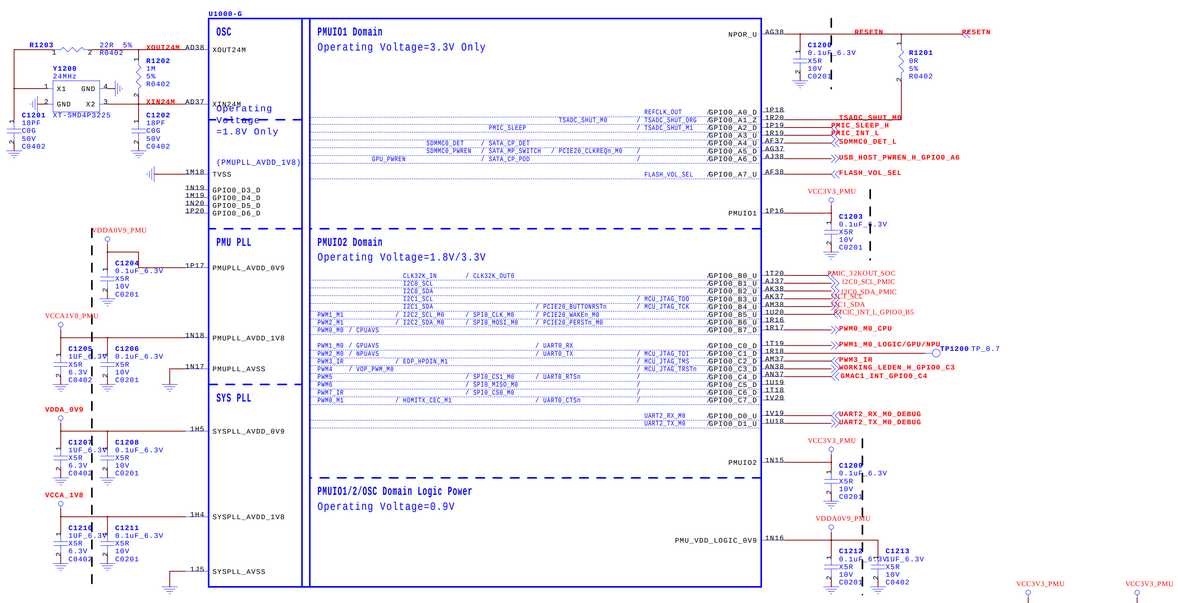
RK3566_F(DDR PHY)



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RK3566_G(OSC/PLL/PMUIO1/2)



Note:

Caps of between dashed green lines and U1000 ~~should be placed under the U1000 package.~~
Other caps should be placed close to the U1000 package

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RK3566_I(VCCIO2 Domain)

U1000-I

VCCIO2 Domain

Operating Voltage=1.8V/3.3V

EMMC_D0	/ FLASH_D0	/GPIO1_B4_U	A32	EMMC_D0/FLASH_D0
EMMC_D1	/ FLASH_D1	/GPIO1_B5_U	B27	EMMC_D1/FLASH_D1
EMMC_D2	/ FLASH_D2	/GPIO1_B6_U	B32	EMMC_D2/FLASH_D2
EMMC_D3	/ FLASH_D3	/GPIO1_B7_U	B29	EMMC_D3/FLASH_D3
EMMC_D4	/ FLASH_D4	/GPIO1_C0_U	B33	EMMC_D4/FLASH_D4
EMMC_D5	/ FLASH_D5	/GPIO1_C1_U	A30	EMMC_D5/FLASH_D5
EMMC_D6	/ FLASH_D6	/GPIO1_C2_U	B30	EMMC_D6/FLASH_D6
EMMC_D7	/ FLASH_D7	/GPIO1_C3_U	A33	EMMC_D7/FLASH_D7
EMMC_CMD	/ FLASH_WRn	/GPIO1_C4_U	A27	EMMC_CMD/FLASH_WRN
EMMC_CLKOUT	/ FLASH_DQS	/GPIO1_C5_D	A29	EMMC_CLKOUT/FLASH_DQS_SOC
EMMC_DATA_STROBE	/ FSPI_CS1n / FLASH_CLE	/GPIO1_C6_D	1A16	EMMC_DATA_STROBE/FLASH_CLE
EMMC_RSTn	/ FSPI_D2 / FLASH_WPn	/GPIO1_C7_D	1B16	EMMC_RSTN/FLASH_WPN
	FSPI_CLK / FLASH_ALE	/GPIO1_D0_D	1A15	
	FSPI_D0 / FLASH_RDY	/GPIO1_D1_U	1A12	
	FSPI_D1 / FLASH_REn	/GPIO1_D2_U	1A18	
	FSPI_D3 / FLASH_CS0n	/GPIO1_D3_U	1B17	
	FLASH_CS1n	/GPIO1_D4_U	1C15	

Default is determined by FLASH_VOL_SEL/GPIO1_A0_U

L:VCCIO2 must supply 3.3V

H:VCCIO2 must supply 1.8V

VCCIO2 1C13

R1300 22R 5% R0201 (EMMC_CLKOUT/FLASH_DQS_SOC → EMMC_CLKOUT/FLASH_DQS)

VCCIO_FLASH

C1300 0.1uF_6.3V X5R 10V C0201

J75

EMMC_D0/FLASH_D0 — TEST08 2T6 — TEST08

Note:
Reserve TestPoint for put the system into Maskrom mode to update the firmware

RK3566_J(VCCIO3 Domain)

U1000-J

VCCIO3 Domain

Operating Voltage=1.8V/3.3V

PWM8_M1	/ SDMMC0_D0 / UART2_TX_M1 / UART6_TX_M1	/GPIO1_D5_U	1E20	SDMMC0_D0
PWM9_M1	/ SDMMC0_D1 / UART2_RX_M1 / UART6_RX_M1	/GPIO1_D6_U	1E19	SDMMC0_D1
	SDMMC0_D2 / ARM_JTAG_TCK/ UART5_CTSn_M0	/GPIO1_D7_U	1D20	SDMMC0_D2/ARMJTAG_TCK
	SDMMC0_D3 / ARM_JTAG_TMS/ UART5_RTSn_M0	/GPIO2_A0_U	1E18	SDMMC0_D3/ARMJTAG_TMS
PWM10_M1	/ SDMMC0_CMD / UART5_RX_M0	/GPIO2_A1_U	1E19	SDMMC0_CMD
	SDMMC0_CLK / TEST_CLKOUT / UART5_TX_M0	/GPIO2_A2_D	G38	SDMMC0_CLK_SOC → SDMMC0_CLK

R1301 22R 5% R0402

VCCIO_SD

VCCIO3 1F17

C1301 0.1uF_6.3V X5R 10V C0201

Note:

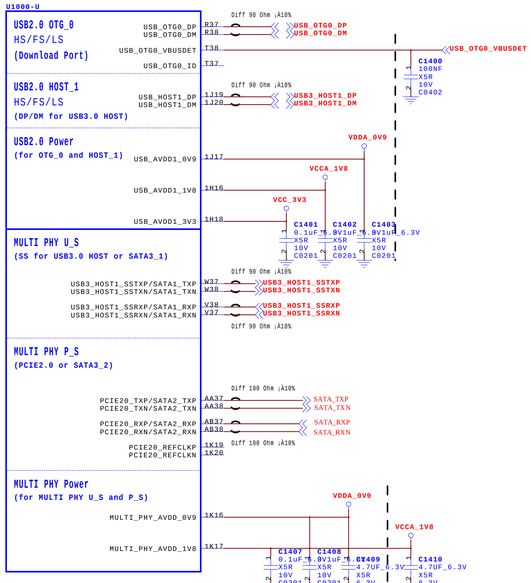
Caps of between dashed green lines and U1000 should be placed under the U1000 package

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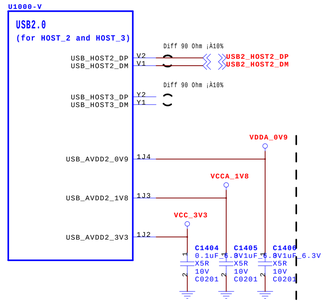
RK3566_U(USB3.0/PCIe2.0 x1)



Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

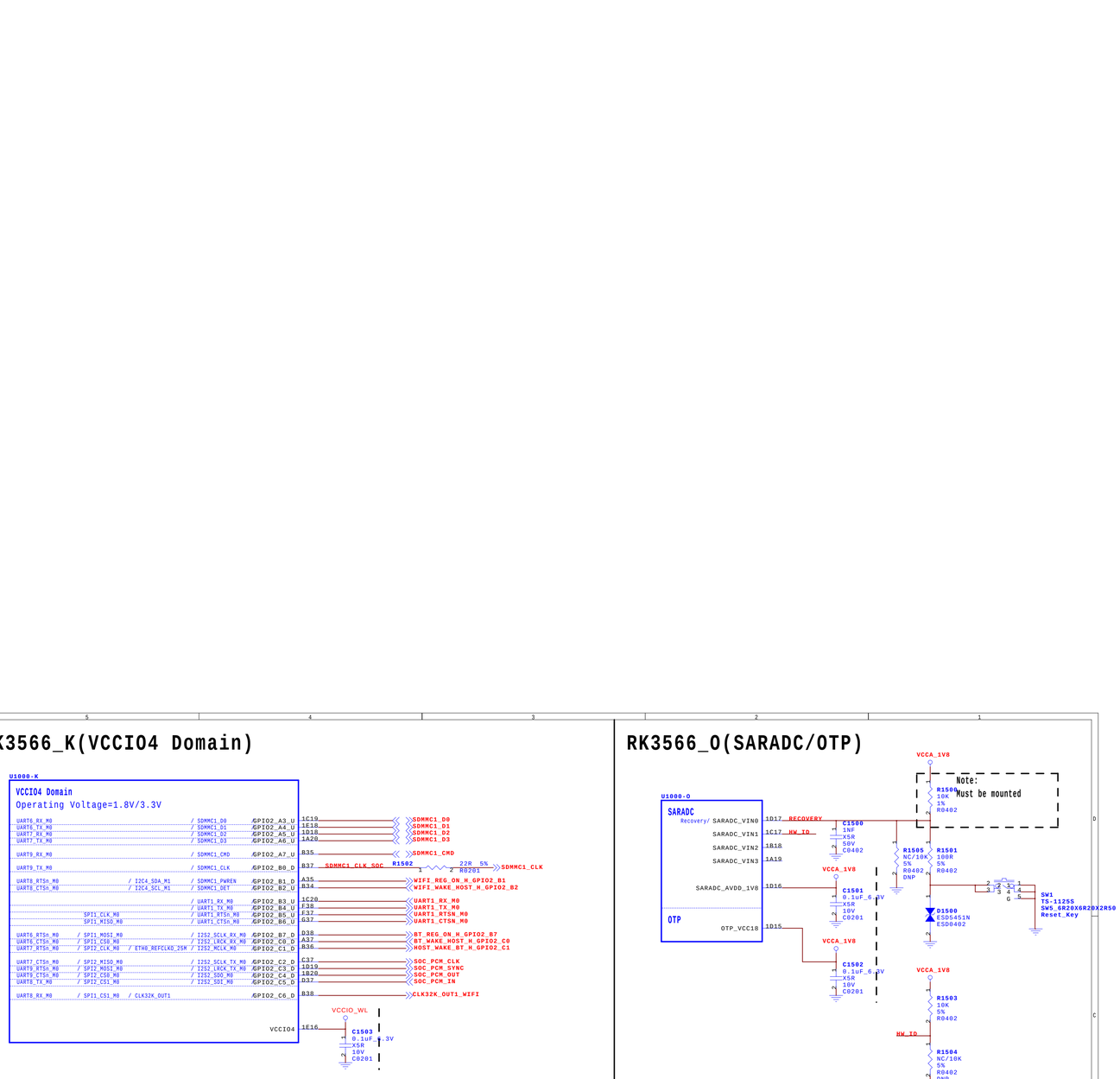
RK3566_V(USB2.0 HOST)



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RK3566_K(VCCIO4 Domain)



RK3566_O(SARADC/OTP)

RK3566_N(VCCIO7 Domain)

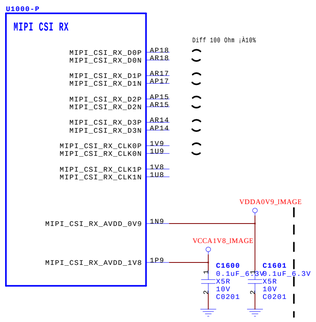
Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package.

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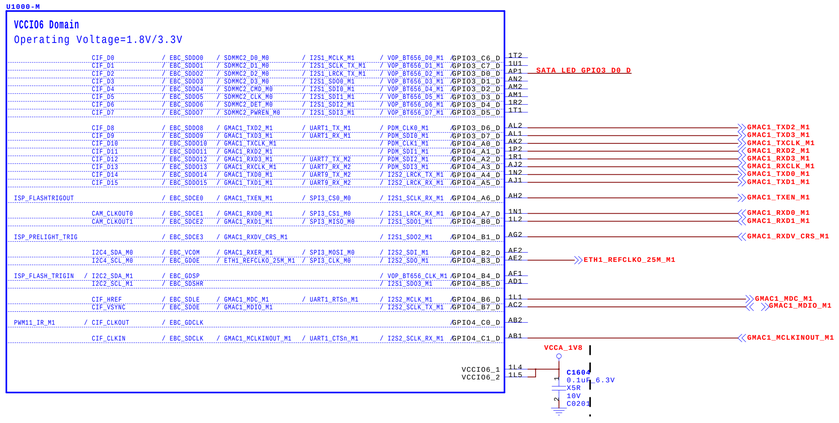
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RK3566_P(MIPI_CSI_RX)



Option1	Sensor1 x4Lane	MIPI_CSI_RX_D0-3 MIPI_CSI_RX_CLK0
Option2	Sensor1 x2Lane + Sensor2 x2Lane	MIPI_CSI_RX_D0-1 MIPI_CSI_RX_CLK0 MIPI_CSI_RX_D2-3 MIPI_CSI_RX_CLK1

RK3566_M(VCCIO6 Domain)



Mode	16bit	12bit	10bit	8bit
CIF_D0	D0	--	--	--
CIF_D1	D1	--	--	--
CIF_D2	D2	--	--	--
CIF_D3	D3	--	--	--
CIF_D4	D4	D0	--	--
CIF_D5	D5	D1	--	--
CIF_D6	D6	D2	D0	--
CIF_D7	D7	D3	D1	--
CIF_D8	D8	D4	D2	D0
CIF_D9	D9	D5	D3	D1
CIF_D10	D10	D6	D4	D2
CIF_D11	D11	D7	D5	D3
CIF_D12	D12	D8	D6	D4
CIF_D13	D13	D9	D7	D5
CIF_D14	D14	D10	D8	D6
CIF_D15	D15	D11	D9	D7

Support BT601 YCbCr 422 8bit input
Support BT656 YCbCr 422 8bit input
Support RAW 8/10/12bit input
Support BT1120 YCbCr 422 8/10/12/16bit input, single/dual-edge sampling
Support 2/4 mixed BT656/BT1120 YCbCr 422 8bit input

Note:

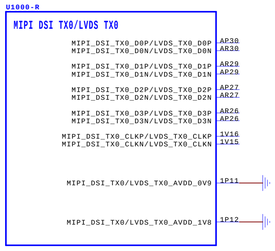
Caps of between dashed green lines and U1000 ~~should be placed under the U1000 package~~.
Other caps should be placed close to the U1000 package

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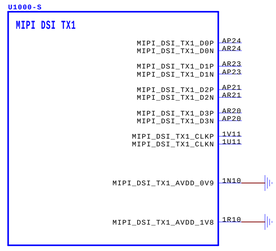
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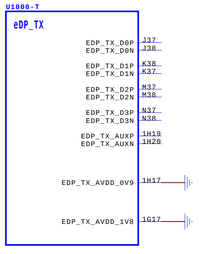
RK3566_R(MIPI_DSI_TX0/LVDS_TX0)



RK3566_S(MIPI_DSI_TX1)



RK3566_T(eDP TX)

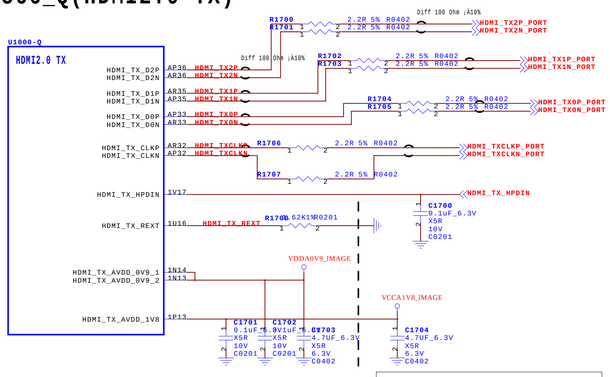


Note:

Caps of between dashed green lines and U1000 ~~should be placed under the U1000 package~~.

Other caps should be placed close to the U1000 package

RK3566_Q(HDMI2.0 TX)



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RK3566_L(VCCIO5 Domain)

U1000-L

VCCIO5 Domain

Operating Voltage=1.8V/3.3V

	VOP_BT1120_D0	/ SPI1_CS0_M1			/ SDMMC2_D0_M1	/GPIO3_A1_D	1U5
	VOP_BT1120_D1		/ GMAC1_TXD2_M0	/ I2S3_MCLK_M0	/ SDMMC2_D1_M1	/GPIO3_A2_D	AR10
	VOP_BT1120_D2		/ GMAC1_TXD3_M0	/ I2S3_SCLK_M0	/ SDMMC2_D2_M1	/GPIO3_A3_D	AP10
	VOP_BT1120_D3		/ GMAC1_RXD2_M0	/ I2S3_LRCK_M0	/ SDMMC2_D3_M1	/GPIO3_A4_D	AR9
	VOP_BT1120_D4		/ GMAC1_RXD3_M0	/ I2S3_SDO_M0	/ SDMMC2_CMD_M1	/GPIO3_A5_D	AP9
	VOP_BT1120_CLK		/ GMAC1_TXCLK_M0	/ I2S3_SDI_M0	/ SDMMC2_CLK_M1	/GPIO3_A6_D	1U4
	VOP_BT1120_D5		/ GMAC1_RXCLK_M0		/ SDMMC2_DET_M1	/GPIO3_A7_D	1V3
	VOP_BT1120_D6		/ ETH1_REFCLKO_25M_M0		/ SDMMC2_PWREN_M1	/GPIO3_B0_D	AR7
PWM8_M0	/ VOP_BT1120_D7		/ GMAC1_RXD0_M0	/ UART4_RX_M1		/GPIO3_B1_D	AP7
PWM9_M0	/ VOP_BT1120_D8		/ GMAC1_RXD1_M0	/ UART4_TX_M1		/GPIO3_B2_D	AR6
	VOP_BT1120_D9	/ I2C5_SCL_M0	/ GMAC1_RXDV_CRS_M0		/ PDM_SDI0_M2	/GPIO3_B3_D	AP6
	VOP_BT1120_D10	/ I2C5_SDA_M0	/ GMAC1_RXER_M0		/ PDM_SDI1_M2	/GPIO3_B4_D	1U3
PWM10_M0	/ VOP_BT1120_D11	/ I2C3_SCL_M1	/ GMAC1_TXD0_M0			/GPIO3_B5_D	1T4
PWM11_IR_M0	/ VOP_BT1120_D12	/ I2C3_SDA_M1	/ GMAC1_TXD1_M0			/GPIO3_B6_D	1V2
PWM12_M0			/ GMAC1_TXEN_M0	/ UART3_TX_M1	/ PDM_SDI2_M2	/GPIO3_B7_D	AP5
PWM13_M0			/ GMAC1_MCLKINOUT_M0	/ UART3_RX_M1	/ PDM_SDI3_M2	/GPIO3_C0_D	AR4
	VOP_BT1120_D13	/ SPI1_MOSI_M1		/ PCIE20_PERSTn_M1	/ I2S1_SDO2_M2	/GPIO3_C1_D	AP4
	VOP_BT1120_D14	/ SPI1_MISO_M1		/ UART5_TX_M1	/ I2S1_SDO3_M2	/GPIO3_C2_D	AP3
	VOP_BT1120_D15	/ SPI1_CLK_M1		/ UART5_RX_M1	/ I2S1_SCLK_RX_M2	/GPIO3_C3_D	AR2
PWM14_M0		/ VOP_PWM_M1	/ GMAC1_MDC_M0	/ UART7_TX_M1	/ PDM_CLK1_M2	/GPIO3_C4_D	1P3
PWM15_IR_M0		/ SPDIF_TX_M1	/ GMAC1_MDIO_M0	/ UART7_RX_M1	/ I2S1_LRCK_RX_M2	/GPIO3_C5_D	1P4
						VCCIO5_1	1N5
						VCCIO5_2	1N6

Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

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RK3566_H(VCCIO1 Domain)

U1000-H

VCCIO1 Domain

Operating Voltage=1.8V/3.3V

	/ I2C3_SDA_M0	/ UART3_RX_M0	/ AUDIOPWM_LOUT_p	GPIO1_A0_U	A22
	/ I2C3_SCL_M0	/ UART3_TX_M0	/ AUDIOPWM_LOUT_n	GPIO1_A1_U	B22
SCR_CLK	/ I2S1_MCLK_M0	/ UART3_RTSn_M0		GPIO1_A2_D	A23
SCR_IO	/ I2S1_SCLK_TX_M0	/ UART3_CTSn_M0		GPIO1_A3_D	B23
	I2S1_SCLK_RX_M0	/ UART4_RX_M0	/ PDM_CLK1_M0 / SPDIF_TX_M0	GPIO1_A4_D	1A13
SCR_RST	/ I2S1_LRCK_TX_M0	/ UART4_RTSn_M0		GPIO1_A5_D	B24
	I2S1_LRCK_RX_M0	/ UART4_TX_M0	/ PDM_CLK0_M0 / AUDIOPWM_ROUT_p	GPIO1_A6_D	1A14
SCR_DET	/ I2S1_SDO0_M0	/ UART4_CTSn_M0	/ AUDIOPWM_ROUT_n	GPIO1_A7_D	B25
	I2S1_SDO1_M0	/ I2S1_SDI3_M0	/ PDM_SDI3_M0 / PCIE20_CLKREQn_M2	GPIO1_B0_D	1B13
	I2S1_SDO2_M0	/ I2S1_SDI2_M0	/ PDM_SDI2_M0 / PCIE20_WAKEn_M2	GPIO1_B1_D	A26
	I2S1_SDO3_M0	/ I2S1_SDI1_M0	/ PDM_SDI1_M0 / PCIE20_PERSTn_M2	GPIO1_B2_D	B26
		I2S1_SDI0_M0	/ PDM_SDI0_M0	GPIO1_B3_D	1B14
				VCCIO1	1D13

RTC_SCL
RTC_SDA
I2S1_MCLK_M0_RK809
I2S1_SCLK_TX_M0_RK809
SPDIF_TX_M0
PDM_CLK0_M0_RK809
I2S1_LRCK_TX_M0_RK809
I2S1_SDO0_M0_RK809
I2S1_SDI0_M0/PDM_SDI0_M0_RK809

VCCIO_ACODEC

C1000
0.1uF
6.3V
X5R
10V
C0201

Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

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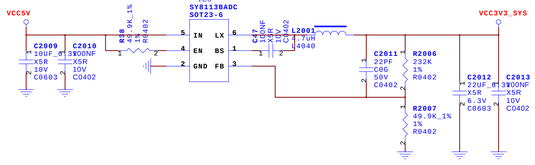
12V/2A DCIN

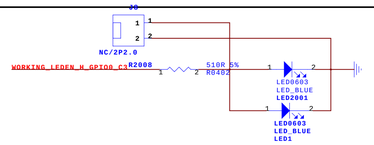
Default 12V

Default 5.18V

VCC3V3_SYS

Default 3.39V





WORKING_LEDEN_H_GPIO0_C3

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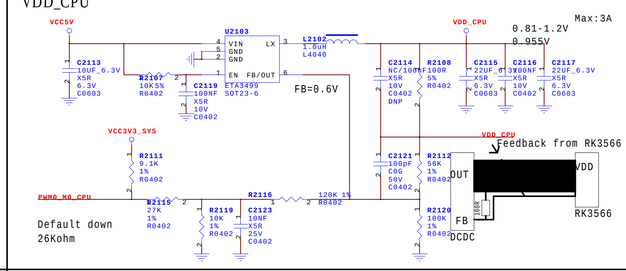
PWM0_M0_CPU

PWM1_M0_LOGIC/GPU/NPU

DDR3=1.5V

VCC_DDR

VDD_CPU



Max:3A

0.81-1.2V

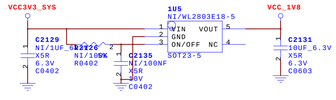
0.955V

Feedback from RK3566

Default down
26Kohm

VCC_3V3

VCC_1V8



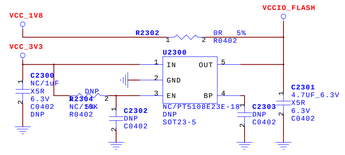
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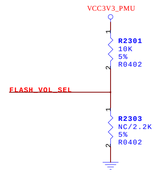
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FLASH_VOL_SEL

Flash Power Manage

	VCCIO2 domain voltage: Recommend voltage value (VCCIO_FLASH)	FLASH_VOL_SEL state decided to VCCIO2 domain IO driven by default
eMMC	1.8V	FLASH_VOL_SEL --> Logic=H
Nand flash	Default 3.3V, Adjust according to demand 1.8V	FLASH_VOL_SEL --> Logic=L(Default)
SPI flash	Default 3.3V, Adjust according to demand 1.8V	FLASH_VOL_SEL --> Logic=L(Default)



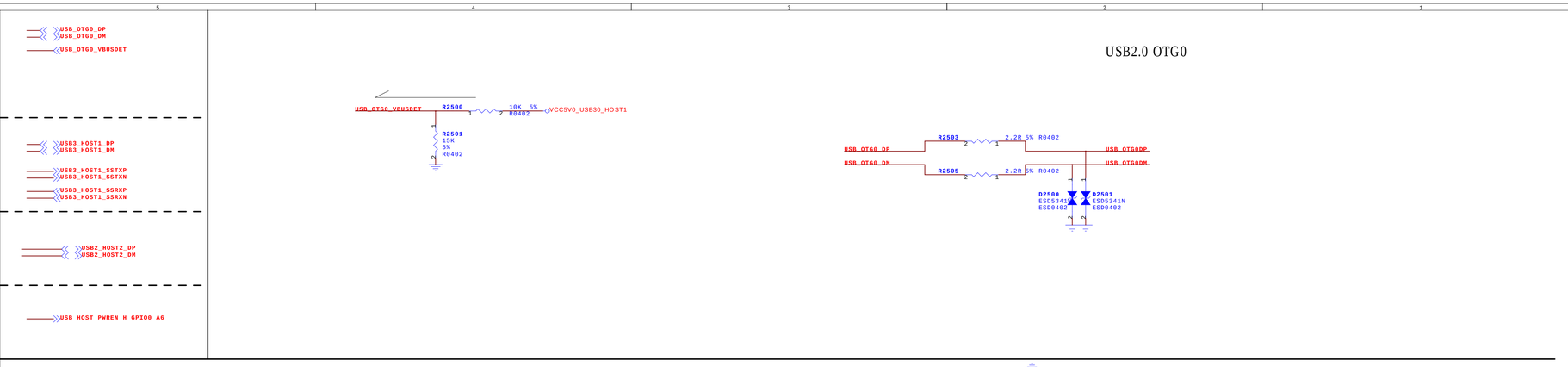


Note:
FLASH_VOL_SEL state decided
to VCCIO2 domain IO driven by default
Logic=LÉ°3.3V IO driven
Logic=HÉ°1.8V IO driven

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USB2.0 OTG0



SATA

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DDR3_DQ0 10
DDR3_DQ1 10
DDR3_DQ2 10
DDR3_DQ3 10
DDR3_DQ4 10
DDR3_DQ5 10
DDR3_DQ6 10
DDR3_DQ7 10
DDR3_DM0 10
DDR3_DQS0N 10
DDR3_DQS0P 10

DDR3_CLKN 10
DDR3_CLKP 10

DDR3_BA2 10
DDR3_BA1 10
DDR3_BA0 10

DDR3_A15 10
DDR3_A14 10
DDR3_A13 10
DDR3_A12 10
DDR3_A11 10
DDR3_A10 10
DDR3_A9 10
DDR3_A8 10
DDR3_A7 10
DDR3_A6 10
DDR3_A5 10
DDR3_A4 10
DDR3_A3 10
DDR3_A2 10
DDR3_A1 10
DDR3_A0 10

DDR3_RESETN 10

DDRPHY_VREFOUT 10

DDR3_DQ8 10
DDR3_DQ9 10
DDR3_DQ10 10
DDR3_DQ11 10
DDR3_DQ12 10
DDR3_DQ13 10
DDR3_DQ14 10
DDR3_DQ15 10
DDR3_DM1 10
DDR3_DQS1N 10
DDR3_DQS1P 10

DDR3_DQ16 10
DDR3_DQ17 10
DDR3_DQ18 10
DDR3_DQ19 10
DDR3_DQ20 10
DDR3_DQ21 10
DDR3_DQ22 10
DDR3_DQ23 10
DDR3_DM2 10
DDR3_DQS2N 10
DDR3_DQS2P 10

DDR3_DQ24 10
DDR3_DQ25 10
DDR3_DQ26 10
DDR3_DQ27 10
DDR3_DQ28 10
DDR3_DQ29 10
DDR3_DQ30 10
DDR3_DQ31 10
DDR3_DM3 10
DDR3_DQS3N 10
DDR3_DQS3P 10

DDR3_RESETN
DDR3_CSON
DDR3_CS1N
DDR3_WEN
DDR3_CKE
DDR3_ODT0
DDR3_ODT1
DDR3_CS0N
DDR3_CS1N
DDR3_CLKN
DDR3_CLKP
DDR3_BA2
DDR3_BA1
DDR3_BA0
DDR3_A15
DDR3_A14
DDR3_A13
DDR3_A12
DDR3_A11
DDR3_A10
DDR3_A9
DDR3_A8
DDR3_A7
DDR3_A6
DDR3_A5
DDR3_A4
DDR3_A3
DDR3_A2
DDR3_A1
DDR3_A0
DDR3_RASN
DDR3_CASN
DDR3_WEN

VCC_DDR
VCC5V
DDR_VTT
DDRPHY_VREFOUT
VREF_DDR_CA
VREF_DDR_DQ
GND

U2 U3 U4 U5 U6 U7 U8
K4B4G0846D-BCMA
BGA78-0P8MM

3C1 3C2 3C3 3C4 3C5 3C6 3C7 3C8 3C9 3C10 3C11 3C12 3C13 3C14 3C15 3C16 3C17 3C18
1UF_6.3V 100NF 1UF_6.3V 100NF 1UF_6.3V 100NF 1UF_6.3V 1NF 10UF_6.3V
C0402 C0402 C0402 C0402 C0402 C0402 C0402 C0402 C0603

3C19 330PF C0402
3C20 12PF C0402
3C21 12PF C0402
3C24 100PF C0402
3C25 100PF C0402
EMI Option

3R3 3R4 240R_1% R0402

3C22 3C23 100NF C0402

3C26 3C27 100NF C0402

3C28 1NF C0402

R3104 1K 1% R0402
R3106 1K 1% R0402
C3105 1NF C0402 X5R 50V
R3108 DNP NC 1% R0402
R3110 NC 1% R0402
C3127 1NF C0402 X5R 50V
C3149 1NF X5R 50V C0402
C3159 1NF X5R 50V C0402
R3109 0R 5% R0402

Notes:
Close to SOC

U1
APL5337/RT9199PSP/RT9045
APL5337/RCT3101S/UR6517/UR6515C/RT9173CPSP/RT9173DPSP
SOP-9
VIN NC8 GND NC7 REFEN VCNTL VOUT NC5 EPAD
3C52 10UF_6.3V 3C53 2200NF C0603 C0402
3C54 10UF_6.3V 3C55 2200NF C0603 C0402
3C56 100NF C0402
3R9 100K_1% R0402
3R10 100K_1% R0402

3C57 1UF_6.3V C0402
3C58 1UF_6.3V 3C59 100NF C0402
3C60 100NF C0402
3C61 100NF C0402
3C63 1UF_6.3V C0402
3C64 10UF_6.3V C0603
3C65 10UF_6.3V C0603
3C66 1UF_6.3V 3C67 100NF C0402
3C68 100NF C0402
3C69 1UF_6.3V C0402
3C70 100NF C0402
3C71 1UF_6.3V C0402
3C72 100NF C0402
3C73 100NF C0402

3C29 10UF_6.3V C0603
3C30 3C31 100NF C0402
3C32 100NF C0402
3C33 1UF_6.3V C0402
3C34 100NF C0402
3C35 1UF_6.3V C0402
3C36 10UF_6.3V C0603
3C37 10UF_6.3V C0603
3C38 10UF_6.3V C0603
3C39 1UF_6.3V C0402
3C40 100NF C0402
3C41 100NF C0402
3C42 1UF_6.3V C0402
3C43 100NF C0402
3C44 100NF C0402
3C45 100NF C0402
3C46 1NF C0402
3C47 10UF_6.3V C0603

3C48 100NF C0402
3C49 100NF C0402
3C50 100NF C0402
3C51 100NF C0402

3C110 10UF_6.3V C0603
3C111 3C112 100NF C0402
3C113 100NF C0402
3C117 10UF_6.3V C0603
3C118 10UF_6.3V C0603
3C119 10UF_6.3V C0603
3C128 10UF_6.3V C0603

3C137 100NF C0402
3C138 100NF C0402
3C139 100NF C0402
3C140 100NF C0402

3C79 330PF C0402
3C97 100PF C0402

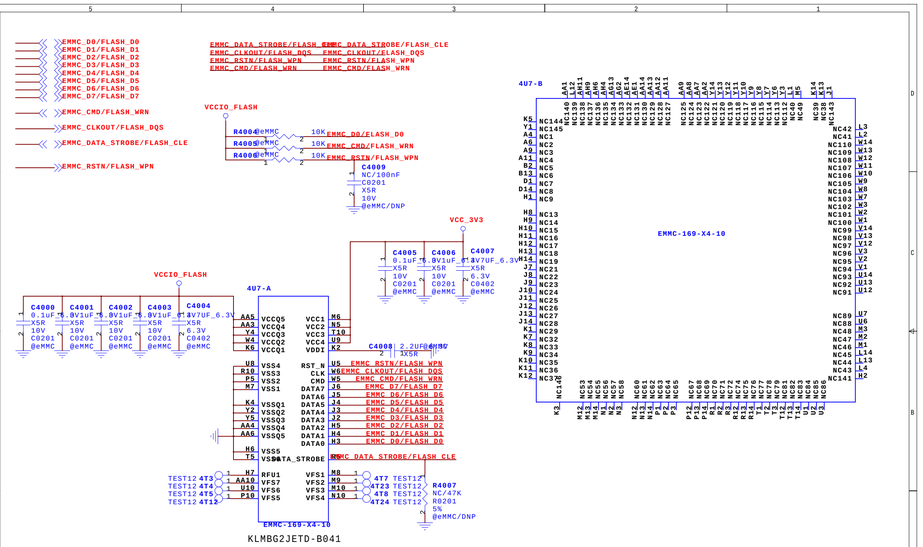
3R7 3R8 3R11 3R12 3R19 3R20 240R_1% R0402

RP1 RP2 RP3 RP4 RP5 RP6 RP7 RP8 RP9 RP10 RP11 RP12 RP13 RP14
4*100R R0402
DDR3_A8 DDR3_A9 DDR3_A6 DDR3_A5 DDR3_CLKP DDR3_CLKN DDR3_CASN DDR3_A11 DDR3_A13 DDR3_A14 DDR3_A12 DDR3_BA1 DDR3_BA2 DDR3_A0 DDR3_A10 DDR3_A4 DDR3_A7 DDR3_A3 DDR3_A1 DDR3_RASN DDR3_ODT0 DDR3_CSON DDR3_A15 DDR3_CLKN DDR3_CKE DDR3_CS1N DDR3_BA0 DDR3_WEN DDR3_A2

DDR_VTT
3R13 100R R0402 DDR3_A9
3R14 100R R0402 DDR3_ODT1

3C80 3C81 3C82 3C83 3C84 4.7UF_6.3V 3C85 4.7UF_6.3V 3C88 4.7UF_6.3V 3C89 4.7UF_6.3V 3C90 1UF_6.3V 3C91 1UF_6.3V 3C92 1UF_6.3V 3C93 1UF_6.3V
C0603 C0402

3C100 3C101 3C102 3C103 3C104 3C105 3C106 3C107 3C108 3C109 3C129 3C130 3C131 3C132 3C133 3C134 3C135 3C136
100NF C0402
1NF C0402



If Flash is compatible, please notice
when eMMC is used, the option is that @eMMC is mounted, @Nand is not mounted,@SPI Flash is not mounted
when Nand is used, the option is that @Nand is mounted, @eMMC is not mounted,@SPI Flash is not mounted
when SPI Flash is used, the option is that SPI Flash is mounted, @eMMC is not mounted,@Nand is not mounted

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SDMMC0_D0
SDMMC0_D1
SDMMC0_D2/ARMJTAG_TCK
SDMMC0_D3/ARMJTAG_TMS
SDMMC0_CMD
SDMMC0_CLK
SDMMC0_DET_L

MicroSD Card

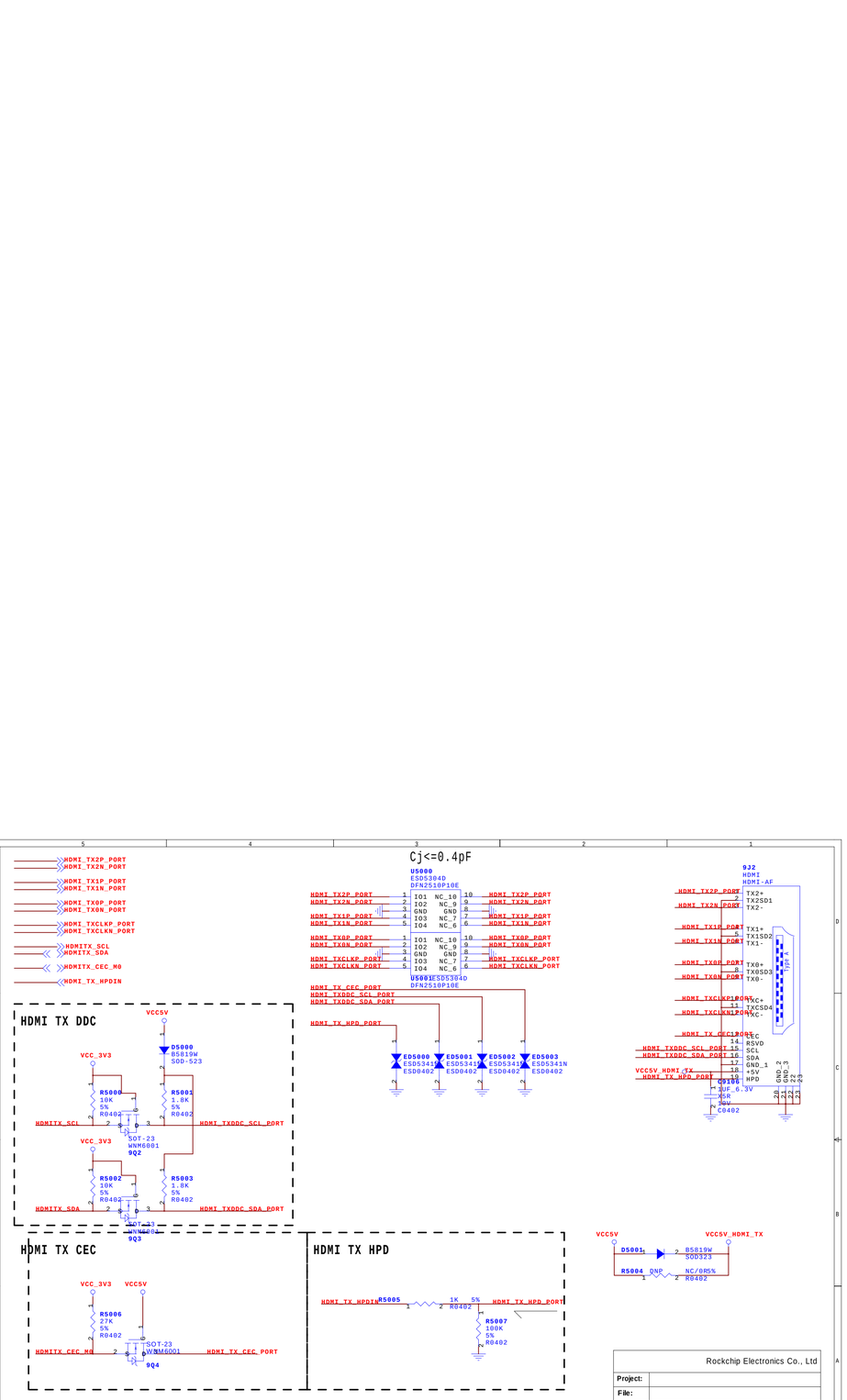
SDMMC0_D2/ARMJTAG_TCK
SDMMC0_D3/ARMJTAG_TMS
SDMMC0_CMD
SDMMC0_CLK
SDMMC0_D0
SDMMC0_D1
SDMMC0_DET_L

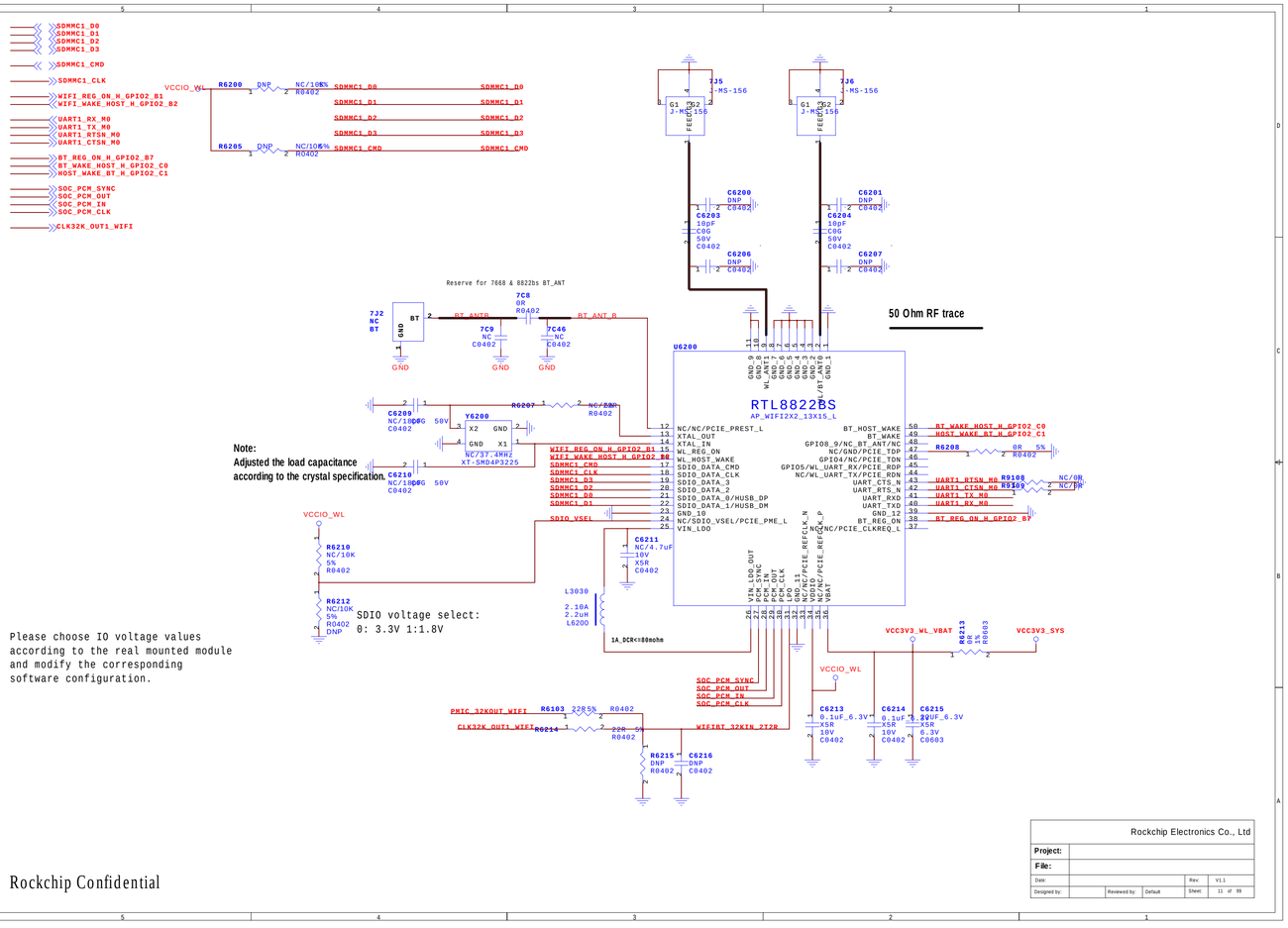
Close to MicroSD Card

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Cj<=0.4pF
HDMI_TX2P_PORT
HDMI_TX2N_PORT
HDMI_TX1P_PORT
HDMI_TX1N_PORT
HDMI_TX0P_PORT
HDMI_TX0N_PORT
HDMI_TXCLKP_PORT
HDMI_TXCLKN_PORT
HDMITX_SCL
HDMITX_SDA
HDMITX_CEC_M0
HDMI_TX_HPDIN
U5000
ESD5304D
DFN2510P10E
U5001
ESD5304D
DFN2510P10E
HDMI_TX_CEC_PORT
HDMI_TXDDC_SCL_PORT
HDMI_TXDDC_SDA_PORT
HDMI_TX_HPD_PORT
ED5000 ESD5341 ESD0402
ED5001 ESD5341 ESD0402
ED5002 ESD5341 ESD0402
ED5003 ESD5341N ESD0402
HDMI TX DDC
VCC5V
VCC_3V3
D5000 B5819W SOD-523
R5000 10K 5% R0402
R5001 1.8K 5% R0402
R5002 10K 5% R0402
R5003 1.8K 5% R0402
SOT-23 WNM6001
Q2
Q3
HDMI TX CEC
R5006 27K 5% R0402
SOT-23 WNM6001
Q4
HDMI TX HPD
R5005 1K 5% R0402
R5007 100K 5% R0402
J2 HDMI HDMI-AF
TX2+ TX2SD1 TX2-
TX1+ TX1SD2 TX1-
TX0+ TX0SD3 TX0-
TXC+ TXSCD4 TXC-
CEC RSVD SCL SDA GND_1 +5V HPD
Type A
GND_2 GND_3 GND_4
C5106 1UF_6.3V X5R 20V C0402
VCC5V_HDMI_TX
HDMI_TX_HPD_PORT
D5001 B5819W SOD323
R5004 DNP NC/0R5% R0402
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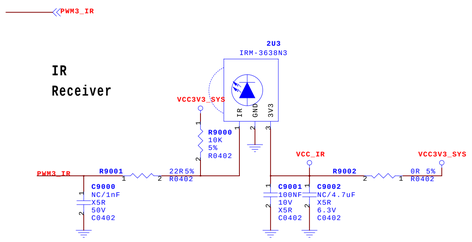
Please choose IO voltage values according to the real mounted module and modify the corresponding software configuration.

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GMAC1_TXD0_M1
GMAC1_TXD1_M1
GMAC1_TXD2_M1
GMAC1_TXD3_M1
GMAC1_TXEN_M1
GMAC1_TXCLK_M1
GMAC1_RXD0_M1
GMAC1_RXD1_M1
GMAC1_RXD2_M1
GMAC1_RXD3_M1
GMAC1_RXDV_CRS_M1
GMAC1_RXCLK_M1
ETH1_REFCLKO_25M_M1
GMAC1_MCLKINOUT_M1
GMAC1_MDC_M1
GMAC1_MDIO_M1
GMAC1_RSTN_GPIO4_C2
GMAC1_INT_GPIO0_C4
Y6800
PHY1_XTALOUT
PHY1_XTALIN
C6800 18pF C0G 50V C0402
C6801 18pF C0G 50V C0402
R6802 NC/0R 5% R0402 DNP
25MHz XT-SMD4P3225
NC/0R
ETH1_REFCLKO_25M_M1
R6811 DNP 5% R0402
PHY1_XTALOUT
GMAC1_MCLKINOUT_M1
R6814 100R 5% R0402
PHY1_CLKOUT32
C6809 DNP C0402
R6815 120R 5% R0402
3.3Vpp
MAC <----- PHY
5T1
G11T1001
T-24P-PT50MIL
PHY1_MDI0+
PHY1_MDI0-
PHY1_MDI1+
PHY1_MDI1-
PHY1_MDI2+
PHY1_MDI2-
PHY1_MDI3+
PHY1_MDI3-
SC10 2.2NF 2KV C0402
SC11 2.2NF 2KV C0402
SC12 2.2NF 2KV C0402
SC13 2.2NF 2KV C0402
M-BI_DAP_TXP
M-BI_DAN_TXN
M-BI_DBP_RXP
M-BI_DBN_RXN
M-BI_DCP
M-BI_DCN
M-BI_DDP
M-BI_DDN
SR1 75R_1% R0402
SR2 75R_1% R0402
SR3 75R_1% R0402
SR4 75R_1% R0402
SR5 R0402510R_1%
SR6 R0402510R_1%
SR7 R0402NC/10R
SR8 R0402120R_1%
SR9 R0402NC/0R
PHY1_LED1/CFG_LDO0
PHY1_LED2/CFG_LDO1
VCC3V3_PHY1
GND
5C9 1NF_2KV
RGM_CGND
J5
BI_DAP_TXP
BI_DAN_TXN
BI_DBP_RXP
BI_DCP
BI_DCN
BI_DBN_RXN
BI_DDP
BI_DDN
YLED-
YLED+
GLED-
GLED+
GND
NC
NC
RJ45-56-LED
R6820 2.49K 1% R0402
VDD10_PHY1
VCC3V3_PHY1
R6823 0R 5% R0402
GMAC1_INT_GPIO0_C4
U6801
E_PAD
MDI[0]+
MDI[0]-
AVDD10
MDI[1]+
MDI[1]-
MDI[2]+
MDI[2]-
AVDD10
MDI[3]+
MDI[3]-
AVDD33
RSET
AVDD10
XTAL_OUT/EXT_CLK
XTAL_IN
CLKOUT
LED2/CFG_LDO1
LED1/CFG_LDO0
LED0/EPHYCR0
REG_OUT
DVDD33
DVDD_RG
RXC/PHYAD1
RXCTL/PHYAD2
RXD0/RXDLY
RXD1/TXDLY
RXD2/PLLOFF
RXD3/PHYAD0
DVDD10
PHY1_REG_OUT
VCC3V3_PHY1
VCCIO_PHY1
PHY1_RXCLK/PHYAD1
PHY1_RXDV/PHYAD2
PHY1_RXD0/RXDLY
PHY1_RXD1/TXDLY
PHY1_RXD2/PLLOFF
PHY1_RXD3/PHYAD0
VDD10_PHY1
XVDD33
PHYRSTB
MDC
MDIO
TXD3
TXD2
TXD1
TXD0
TXCTL
TXC
RTL8211F-CG
QFN40-PT04-5X5MM
RTL8211F-CG(SW Mode)
RTL8211FI-CG(SW Mode)Industrial
VCC3V3_PHY1
R6836 DNP NC/4.99K R0402
R6837 1.8K 5% R0402
VCCIO_PHY1
GMAC1_RSTN_GPIO4_C2
R6838 0R 5% R0402
C6820 0.1uF_6.3V X5R 10V C0201
PHYRSTB is 3.3V IO
PHY1_RXD0/RXDLY R6839 22R5% R0402 GMAC1_RXD0_M1
PHY1_RXD1/TXDLY R6840 22R5% R0402 GMAC1_RXD1_M1
PHY1_RXD2/PLLOFF R6841 22R5% R0402 GMAC1_RXD2_M1
PHY1_RXD3/PHYAD0 R6842 22R5% R0402 GMAC1_RXD3_M1
PHY1_RXCLK/PHYAD1 R6843 22R5% R0402 GMAC1_RXCLK_M1
PHY1_RXDV/PHYAD2 R6844 22R5% R0402 GMAC1_RXDV_CRS_M1
C6826 DNP C0201
Close to PHY
R6816 4.7K 5% R0402 PHY1_LED0/CFG_EXT
R6817 DNP NC/4.99K R0402
R6818 4.7K 5% R0402 PHY1_LED2/CFG_LDO1
R6819 4.7K 5% R0402 PHY1_LED1/CFG_LDO0
VCC_PHY0_IO Voltage Config
R6824 4.7K 5% R0402 PHY1_RXCLK/PHYAD1
R6825 DNP NC/4.99K R0402
R6826 4.7K 5% R0402 PHY1_RXDV/PHYAD2
R6821 4.7K 5% R0402 PHY1_RXD3/PHYAD0
PHY Address Config
PHY Address 1 [default] PHYAD[2:0] 3'b001
R6828 4.7K 5% R0402 PHY1_RXD0/RXDLY
R6829 DNP NC/4.99K R0402
Pull-up for additional 2ns delay to RXC for data latching
R6830 4.7K 5% R0402 PHY1_RXD1/TXDLY
R6831 DNP NC/4.99K R0402
Pull-up for additional 2ns delay to TXC for data latching
R6833 DNP 4.7K 5% R0402 PHY1_RXD2/PLLOFF
R6834 4.7K 5% R0402
Pull-up to disable PLL @ ALDPS mode(Low power mode)
VCCIO_PHY1
VCC_1V8
R6832 0R 5% R0402
C6811 1uF_6.3V X5R C0402
C6812 1uF_6.3V X5R C0402
C6813 0.1uF_6.3V X5R 10V C0402
Close to PIN28
VCC_3V3
R6835 0R 5% R0402
C6816 4.7uF_6.3V X5R C0402
C6814 0.1uF_6.3V X5R 10V C0402
C6819 0.1uF_6.3V X5R 10V C0402
C6821 4.7uF_6.3V X5R 6.3V C0402
C6822 0.1uF_6.3V X5R 10V C0402
Close to PIN11,40
Close to PIN29
VDD10_PHY1
PHY1_REG_OUT
L3030 2uH
C6815 4.7uF_6.3V X5R 10V C0603
C6817 0.1uF_6.3V X5R 10V C0402
Close to PIN30
RTL8211F-CG(SW Mode)
RTL8211FI-CG(SW Mode)Industrial
C6818 0.1uF_6.3V X5R 10V C0402
Close to PIN21
C6823 0.1uF_6.3V X5R 10V C0402
C6824 0.1uF_6.3V X5R 10V C0402
C6825 0.1uF_6.3V X5R 10V C0402
Close to PIN3,8,38
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IR Receiver



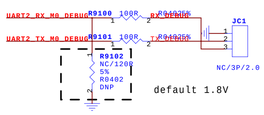
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Debug UART2

UART2_RX_M0_DEBUG
UART2_TX_M0_DEBUG
FP_DATA
FP_CLK
FP_STB

When the project is design,
Please use the circuit below



default 1.8V

Reserved a voltage divide circuit
for 3.3V RX input compatible.

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TOP Mark



BOTTOM Mark



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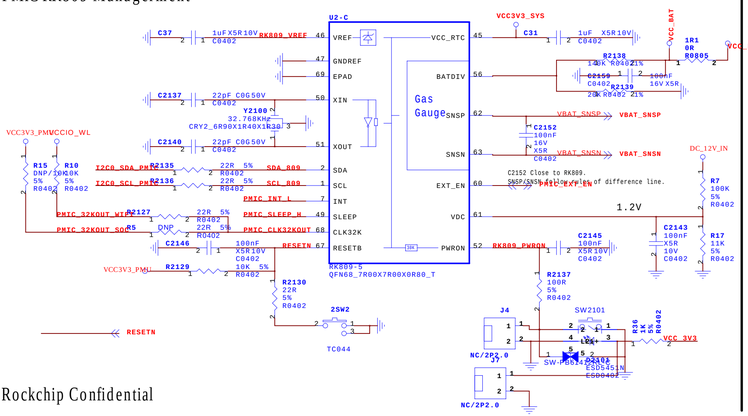
PMIC RK809 DCDC

DDR4	1.2V	62K 1%
LPDDR4/4x	1.1V	47K 1%
DDR3	1.53V	110K 1%
DDR3L	1.35V	82K 1%
LPDDR3	1.2V	62K 1%

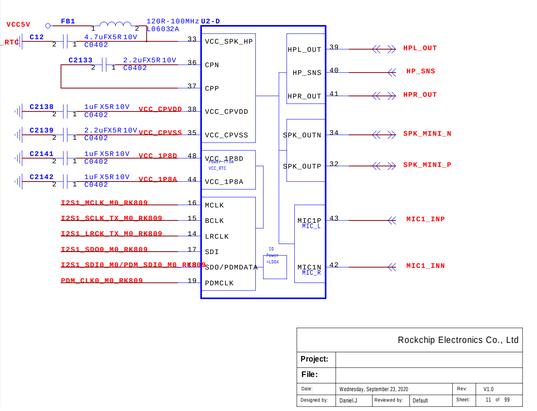
Default:1.1V

PMIC RK809 LDO

PMIC RK809 Managerment



PMIC RK809 CODEC



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I2C0_SCL_PMIC
I2C0_SDA_PMIC
PMIC_SLEEP_H

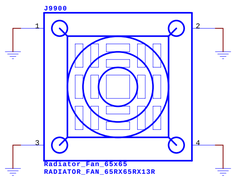
I2C1_SCL
I2C1_SDA
RTCIC_INT_L_GPIO0_B5

RTC IC

²Ï¿¼Í¼ÎÎªBAT54C

Address:Read A3H,Write A2H

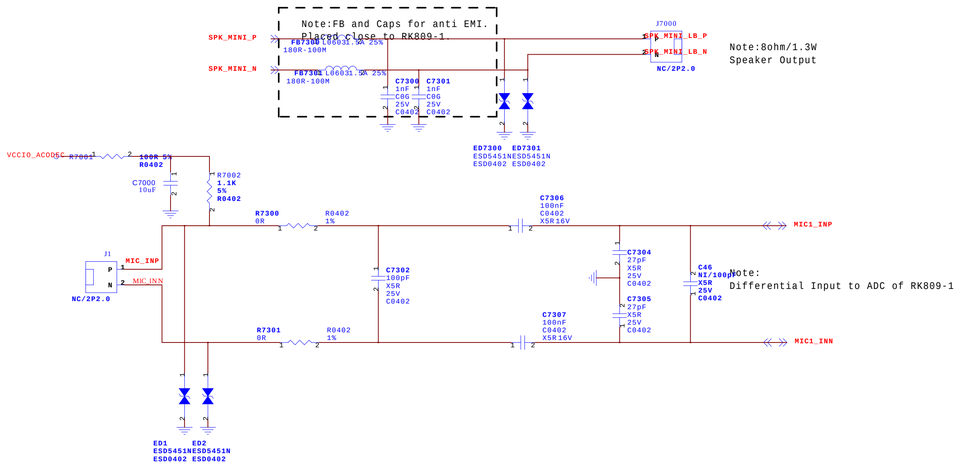
Close to BAT



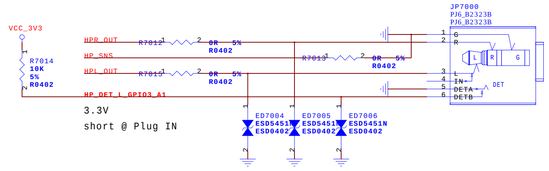
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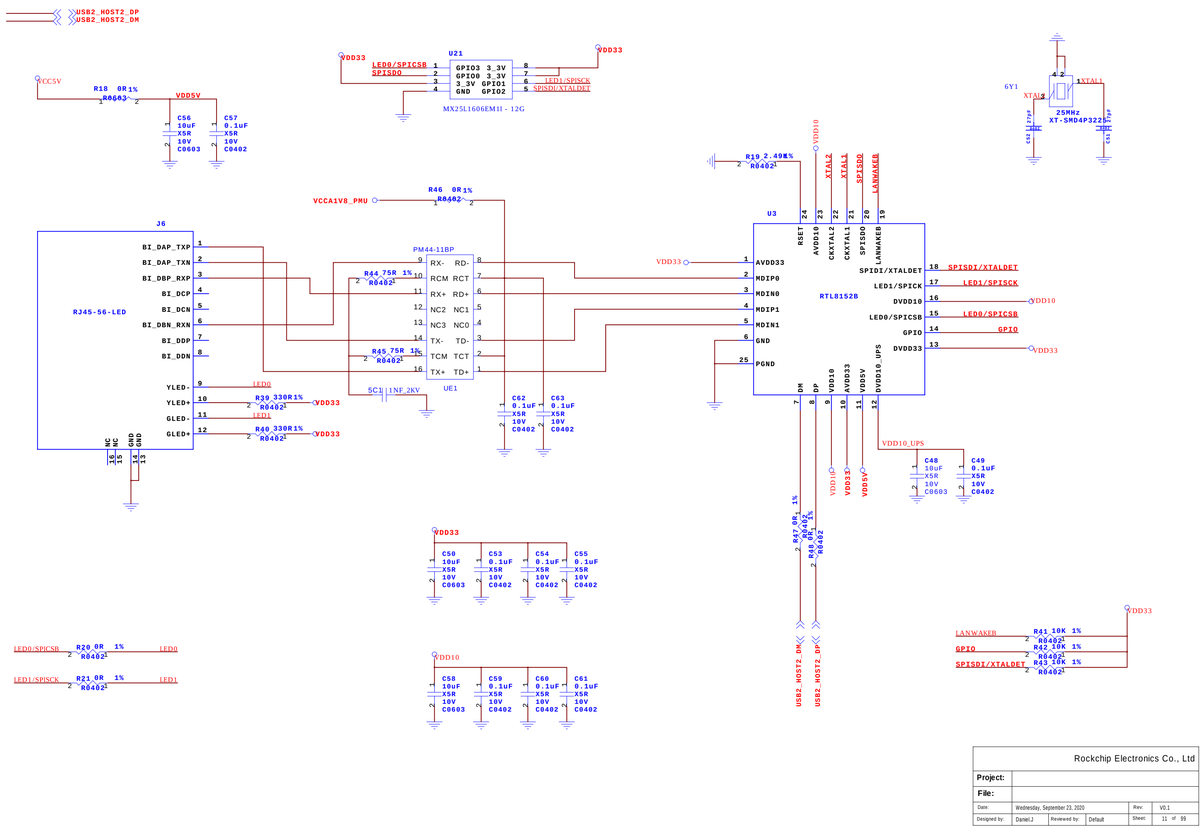
AUDIO_IntegratedAmp



Headphone



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USB2_HOST2_DP
USB2_HOST2_DM
VCC5V
R18 0R 1%
R0603
VDD5V
C56 10uF X5R 10V C0603
C57 0.1uF X5R 10V C0402
VDD33
U21
LED0/SPICSB
SPISDO
GPIO3 3_3V
GPIO0 3_3V
3_3V GPIO1
GND GPIO2
LED1/SPISCK
SPISDI/XTALDET
MX25L1606EM1I - 12G
6Y1
XTAL2
XTAL1
25MHz
XT-SMD4P322
12pf
R19 2.49K 1%
R0402
VDD10
XTAL2
XTAL1
SPISDO
LANWAKEB
U3
RSET
AVDD10
CKXTAL2
CKXTAL1
SPISDO
LANWAKEB
VCCA1V8_PMU
R46 0R 1%
J6
BI_DAP_TXP
BI_DAP_TXN
BI_DBP_RXP
BI_DCP
BI_DCN
BI_DBN_RXN
BI_DDP
BI_DDN
RJ45-56-LED
YLED-
YLED+
GLED-
GLED+
NC
NC
GND
GND
PM44-11BP
RX- RD-
RCM RCT
RX+ RD+
NC2 NC1
NC3 NC0
TX- TD-
TCM TCT
TX+ TD+
R44 75R 1%
R45 75R 1%
R0402
5C1 1NF 2KV
UE1
C62 0.1uF X5R 10V C0402
C63 0.1uF X5R 10V C0402
LED0
LED1
R39 330R 1%
R40 330R 1%
VDD33
AVDD33
MDIP0
MDIN0
MDIP1
MDIN1
GND
PGND
RTL8152B
SPISDI/XTALDET
LED1/SPICK
DVDD10
LED0/SPICSB
GPIO
DVDD33
DVDD10_UPS
DM
DP
VDD10
AVDD33
VDD5V
VDD10_UPS
C48 10uF X5R 10V C0603
C49 0.1uF X5R 10V C0402
R47 0R 1%
R48 0R 1%
USB2_HOST2_DM
USB2_HOST2_DP
C50 10uF X5R 10V C0603
C53 0.1uF X5R 10V C0402
C54 0.1uF X5R 10V C0402
C55 0.1uF X5R 10V C0402
C58 10uF X5R 10V C0603
C59 0.1uF X5R 10V C0402
C60 0.1uF X5R 10V C0402
C61 0.1uF X5R 10V C0402
LED0/SPICSB
R20 0R 1%
LED0
LED1/SPISCK
R21 0R 1%
LED1
LANWAKER
GPIO
SPISDI/XTALDET
R41 10K 1%
R42 10K 1%
R43 10K 1%
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Project:
File:
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Rev: V0.1
Designed by: Daniel.J
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